

Deliverables

1. https://github.com/ArtemisLymperi/seip_assignment_1_2026/
- 2.

Triggered via push 6 minutes ago
ArtemisLymperi pushed · ecf02de main
Status: Success
Total duration: 28s
Artifacts: 1

ci-cd.yaml
on: push

build-and-push 24s

build-and-push summary

Docker Build summary

For a detailed look at the build, download the following build record archive and import it into Docker Desktop's Builds view. Build records include details such as timing, dependencies, results, logs, traces, and other information about a build. [Learn more](#)

ArtemisLymperi-seip_assignment_1_2026-IW46EK.dockerbuild (17.66 KB - includes 1 build record)

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ID	Name	Status	Cached	Duration
IW46EK	https://github.com/ArtemisLymperi/seip_assignment_1_2026.git#ecf02de83392800c408865b64c008e54cfb1a28	completed	0%	8s

► Build inputs

- 3.

```
C:\Users\artem\OneDrive\Εγγράφα\ΟΠΑ\Εξάμηνος\Τεχνολογία Λογισμικού στην Πράξη\seip_assignment_1_2026>kubectl get all -n default
NAME                                READY    STATUS    RESTARTS   AGE
pod/echo-api-deployment-84bddcd9d8-8bn2b  1/1      Running   0           6m44s
pod/echo-api-deployment-84bddcd9d8-h82qj  1/1      Running   0           6m44s
pod/echo-api-deployment-84bddcd9d8-xqwrz  1/1      Running   0           6m44s

NAME                                TYPE          CLUSTER-IP    EXTERNAL-IP  PORT(S)    AGE
service/echo-api-service            ClusterIP     10.101.209.144 <none>       80/TCP     6m44s
service/kubernetes                  ClusterIP     10.96.0.1     <none>       443/TCP    24h

NAME                                READY    UP-TO-DATE    AVAILABLE    AGE
deployment.apps/echo-api-deployment  3/3      3              3            6m44s

NAME                                DESIRED    CURRENT    READY    AGE
replicaset.apps/echo-api-deployment-84bddcd9d8  3          3          3        6m44s

C:\Users\artem\OneDrive\Εγγράφα\ΟΠΑ\Εξάμηνος\Τεχνολογία Λογισμικού στην Πράξη\seip_assignment_1_2026>kubectl get configmap,secret
NAME          DATA  AGE
configmap/echo-api-config  2      11m
configmap/kube-root-ca.crt 1       24h

NAME          TYPE   DATA  AGE
secret/echo-api-secret  Opaque 1      5m11s
```

- 4.

```
C:\Users\artem\OneDrive\Εγγράφα\ΟΠΑ\Εξάμηνος\Τεχνολογία Λογισμικού στην Πράξη\seip_assignment_1_2026>kubectl port-forward service/echo-api-service 8080:80
Forwarding from 127.0.0.1:8080 -> 3000
Forwarding from [::1]:8080 -> 3000
Handling connection for 8080
Handling connection for 8080
```

localhost:8080

Ομορφη εκτύπωση

```
{
  "message": "Welcome to the Software Engineering in Practice Assignment Cluster!",
  "environment": "production"
}
```

localhost:8080/secure-config

Ομορφη εκτύπωση

```
{
  "status": "Authorized",
  "injected_secret_suffix": "*****.key"
}
```

5. AI Reflection & Future Outlook

5.1 AI Integration

I used AI for:

- Identifying errors in the code (YAML indentation, lowercase image tag)
- Formatting the README file
- Providing the correct commands for specific actions (e.g. encoding the API key in Base64)

5.2 Utility Analysis

Aspects of AI that I found most useful:

- Explaining the Base64 encoding and why the `-n` flag is necessary
- Identifying typographical errors in the code
- Understanding how port-forwarding works and how it connects the local machine to the Kubernetes cluster
- Explaining the selector/label relationship between Deployment and Service

5.3 Friction Points

Where the AI tools failed:

- Initially suggested outdated action versions (e.g. actions/checkout@v3). I verified and updated them via the GitHub Marketplace.
- The pipeline failed due to missing write permissions in GitHub Actions. I resolved this manually through the repository Settings (after the error message).
- The AI initially suggested using `echo -n "my-secret-key" | base64` to encode the secret, but this command is not recognized in PowerShell on Windows.

5.4 Future Architectural Outlook

If I had an extra week, the first priority would be replacing `kubectl port-forward` with an Ingress Controller. The current solution is temporary and stops working every time the terminal is closed. An Ingress Controller would expose the application permanently without any manual steps.

After that, I would add image scanning in the CI pipeline to automatically check the Docker image for security issues before pushing it to GHCR.

Prometheus monitoring to keep track of the application's health and performance would be useful but not that applicable within one week.